

Predicting Football Match Outcomes Using Machine Learning and Probability Calibration

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Abstract

Football match results depend on a complex interaction between team performance and competitive dynamics. In this paper, we apply a series of machine learning methods and Isotonic Calibration to predict match outcomes using a dataset of around 230,000 football matches. Bookmaker odds were adjusted with the Newton–Raphson method to account for bookmaker margin effects, while additional calibration procedures were used to improve probability estimation. The proposed machine learning models achieved an AUC of 70.71% in predicting final match outcomes (1-X-2), with particularly strong performance in away-win predictions, reaching an accuracy of 72.02%. Compared to traditional scoring approaches such as Over/Under goals, the proposed method produced more consistent results across different leagues. The analysis also showed that combining statistical match features with calibrated machine learning methods can improve prediction reliability in highly competitive environments.

Keywords: Machine learning; neural networks; stochastic techniques.

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4.1.1. Subsubsection

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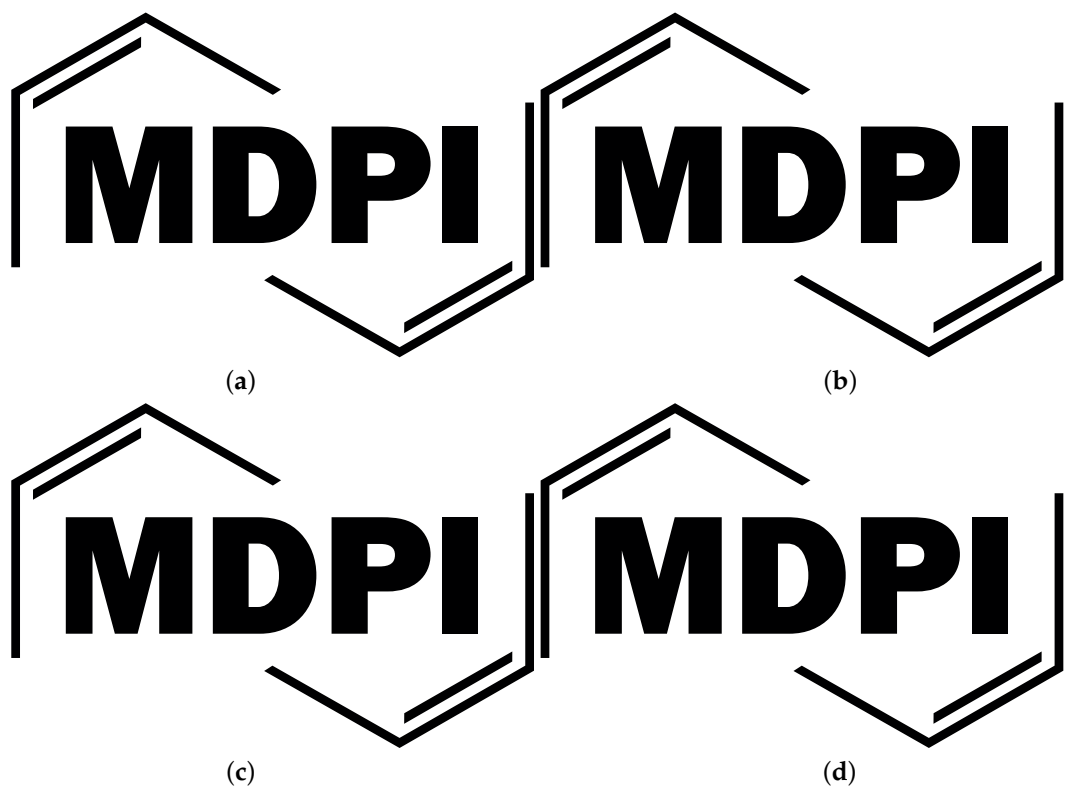


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	Data	Data	Data
	Data	Data	Data
Entry 2	Data	Data	Data
	Data	Data	Data
	Data	Data	Data

* Tables may have a footer.

Text.

Text.

77

78

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79

This is the example 1 of equation:

80

$$a = 1,$$

(1)

81

the text following an equation need not be a new paragraph. Please punctuate equations as regular text.

82

83

This is the example 2 of equation:

84

$$a = b + c + d + e + f + g + h + i + j + k + l + m + n + o + p + q + r + s + t + u + v + w + x + y + z$$

(2)

85

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LD	Linear dichroism

Appendix A

Appendix A.1

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Entry 1	Data	Data
Entry 2	Data	Data

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References

1. Author 1, T. The title of the cited article. *Journal Abbreviation* **2008**, *10*, 142–149.

2. Author 2, L. The title of the cited contribution. In *The Book Title*; Editor1, F., Editor2, A., Eds.; Publishing House: City, Country, 2007; pp. 32–58.

3. Author 1, A.; Author 2, B. *Book Title*, 3rd ed.; Publisher: Publisher Location, Country, 2008; pp. 154–196.

4. Author 1, A.B.; Author 2, C. Title of Unpublished Work. *Abbreviated Journal Name* year, phrase indicating stage of publication (submitted; accepted; in press).

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